Delta Document
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General changes

- Thesis title changed to “Hyperborea: A Generative AI Approach to Snowy Driving Scene Datasets and Benchmarks”.
- Scope changed from “improving perception model performance through training” to evaluating whether generated snowy images are visually realistic and structurally consistent.
- Added --Hyperborea--, a separate 1000-image generated snowy driving dataset from the Berlin, Bielefeld and Munich Cityscapes test subsets.
- Added an abstract and keywords.
- Expanded and cleaned up references.

Abstract

- Added a full abstract summarizing the motivation, method, dataset, evaluation setup and main results.
- Added keywords related to synthetic data generation, diffusion models, ControlNet, Cityscapes, semantic segmentation and adverse weather.

Chapter 1: Introduction

- Rewrote the motivation for snowy driving data and adverse-weather perception.
- Replaced the original research question about improving perception model training with a new research question about visual realism and structural consistency.
- Added three subquestions corresponding to manual evaluation, FID and semantic segmentation.
- Added a short summary of the final results: manual score, FID improvement and segmentation mIoU drop.
- Added/updated the introductory figure showing generated snowy examples.

Chapter 2: Related Work

- Expanded the related work into clearer subsections:

  - Autonomous Driving Datasets
  - Adverse-Weather Datasets
  - Synthetic Data Generation for Autonomous Driving
  - Generative AI
- Added more dataset references, including KITTI, Cityscapes, BDD100K, ACDC, Mapillary Vistas, nuScenes, Waymo Open Dataset, WildDash, CADC, Dark Zurich and Foggy Cityscapes.
- Added earlier synthetic and simulation-based datasets such as SYNTHIA, CARLA, Playing for Data and Synscapes.
- Added image-to-image translation methods such as pix2pix and CycleGAN.
- Added diffusion model background through DDPM, DDIM and Latent Diffusion.

- Added Atlantis as related work because it uses Stable Diffusion to generate synthetic data while preserving scene structure.
- Expanded discussion of WeatherDG, DriveDiTFit, HG-Lane and WeatherDiffusion.
- Clarified how this thesis differs from related work by using a simpler reproducible Stable Diffusion and ControlNet pipeline.

Chapter 3: Problem Definition

- Rewrote the chapter to match the final scope of the thesis.
- Removed the focus on training a perception model.
- Clarified that the main task is to generate snowy versions of clear-weather Cityscapes images while preserving the original scene structure.
- Added a clearer explanation of why annotation reuse depends on structure preservation.
- Added a clearer discussion of the trade-off between snow realism and structural consistency.
- Added explanation of the limitation of Canny edge conditioning: it preserves edges but does not provide semantic understanding.

Chapter 4: Methodology

- Added a methodology pipeline figure.
- Clarified the full pipeline: Cityscapes input, Canny edge extraction, img2img generation with ControlNet, and evaluation.
- Added Canny edge detector citation.
- Expanded data preparation:

  - 267 Frankfurt validation images used for segmentation evaluation.
  - 50 generated images used for manual evaluation.
  - 1000-image Hyperborea dataset generated from Berlin, Bielefeld and Munich test subsets.
- Added a dataset visual examples figure showing original image, ground-truth segmentation and generated snowy image.
- Updated final generation settings:

  - RealisticVision v5.1
  - denoising strength 0.55
  - ControlNet Canny
  - ControlNet weight 0.9
  - final prompt and negative prompt
- Updated batch generation description to match the final one-image-per-input setup.

Chapter 5: Generation Experiments

- Reorganized the old Experiments chapter into a chapter focused only on generation setup.
- Added model comparison between Stable Diffusion 1.4, Stable Diffusion 1.5 and RealisticVision v5.1.
- Added denoising strength discussion and selected 0.55 as final compromise.
- Added prompt development section with final positive and negative prompts.

- Added ControlNet configuration section explaining Canny guidance and ControlNet weight.
- Added final generation configuration section.

Chapter 6: Evaluating the Quality of the Dataset

- Expanded the manual visual evaluation.
- Added a separate scoring criteria section explaining scores 1 to 5.
- Clarified that manual evaluation was performed on 50 generated Frankfurt validation images.
- Added score distribution graph for all evaluation criteria.
- Expanded dataset quality analysis:

  - average overall quality score: 2.98/5
  - 10 images below 2.5
  - 25 images below 3.0
  - 25 images at or above 3.0
- Improved observed strengths and limitations sections.
- Added FID table:

  - Clear Cityscapes vs ACDC Snow: 159.99
  - Generated Snow vs ACDC Snow: 123.87
- Added discussion that FID improved by about 22.6%, but a domain gap remains.

Chapter 7: Semantic Segmentation Benchmark

- Replaced the incomplete Go/No-go segmentation setup.
- Changed from early SegFormer-B0/50-image setup to SegFormer-B5 on 267 Frankfurt validation images.
- Added full evaluation setup:

  - original Frankfurt validation images
  - generated snowy Frankfurt validation images
  - Cityscapes gtFine annotations
  - 19 Cityscapes classes
- Added full per-class IoU table.
- Added mIoU table:

  - clear images: 0.7571
  - generated snowy images: 0.4638
- Added discussion of which classes remain robust and which are strongly affected.
- Added limitations and future work for the segmentation evaluation.

Chapter 8: Conclusion and Discussion

- Replaced the incomplete Go/No-go conclusion with a full conclusion chapter.
- Added sections for:

  - Discussion

  - Answering the Research Question
  - Limitations
  - Future Work
  - Conclusion
- Directly answered the main research question and the three subquestions.
- Added discussion of the main trade-off between snow realism and scene preservation.
- Added Hyperborea as a final thesis output, while clarifying that it was not used for the quantitative evaluation.
- Added clearer limitations and future work directions.

References and formatting

- Expanded bibliography from the Go/No-go version.
- Added citations for datasets, diffusion models, image-to-image translation, ControlNet, Canny, FID and SegFormer.
- Added URL support for software and model repository references.
- Improved tables using clearer formatting.
- Corrected spelling, grammar and terminology throughout the thesis.